INFS1200 Module 4 Assignment

Due: 30 October 2020 @ 11:59 PM AEST

Weighting: 10%

Full Name	Student ID

1. Overview

The purpose of this assignment is to test your ability to use and apply functional dependencies and normalisation in the database design process. You will gain experience in isolating problems with relational schema by determining the appropriate normal form and finding examples of potential anomalies in relation instances. Finally, you will generate an efficient database schema using both the BCNF and 3NF approach to normalisation.

This assignment **must** be completed individually.

2. Submission

All submissions must be made through an electronic marking tool called Gradescope, which will also be used for providing feedback. You **must** record all your answers in the spaces provided in this document. Altering the format or layout of this document in anyway will attract penalties**.**

3. Marking

The Module 4 assignment is worth **10 course marks** (of 40 course marks total for all four assignments).

4. Plagiarism

The University has strict policies regarding plagiarism. Penalties for engaging in unacceptable behaviour can range from cash fines or loss of grades in a course, through to expulsion from UQ. You are required to read and understand the policies on academic integrity and plagiarism in the course profile (Section 6.1).

If you have any questions regarding acceptable level of collaboration with your peers, please see either the lecturer or your tutor for guidance. Remember that ignorance is not a defence!

5. Task

This assignment has been split into five sections which each examine a key topic covered in Module 4 of this course. Each section will have several questions with appropriate instructions to assist you in completing that question. You must complete **all** sections of this assignment and submit your answers using the boxes provided.

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Section A – Anomalies & Functional Dependencies

Question 1

A local grocery shop *Kimchi N' Bread*™ uses a database to store transactions related to customer purchases. A sample of a table in their database has been provided below. For simplicity, the attribute names have been simplified to single letters.

A	B	C	D	E	F
1	Bread	Grace	12/10/2020	4	1000
2	Kimchi	Jack	12/10/2020	1	2000
3	Milk	Grace	14/10/2020	4	2000
4	Bread	Andrew	18/10/2020	8	1000
5	Kimchi	Charlie	18/10/2020	7	2000
6	Milk	Jack	19/10/2020	1	2000

Based on the data above, provide a list of *all possible* non-trivial functional dependences for this table. You do not need to justify your answers.

Note: For this question, non-trivial FD's with more than one attribute on the left-hand side should also not be included. For example, the following two FD's should not be included:

{A} → {A} {A, B} → {A}

Question 2

UQ has decided to create an efficient database to help tutors manage their teaching schedule. The schema for their Tutor table and a sample of the instance data can be seen below.

ID	BuildingNumber	BuildingName	DayTime	TutorNumber	TutorName
10	7	Parnell	Thursday 12:00 PM	1345	Jelena
12	7	Parnell	Thursday 1:00 PM	1345	Jelena
13	47	Axon	Thursday 1:00 PM	4123	Jason
14	68	Chemistry	Friday 08:00 AM	2289	Tahlia
15	68	Chemistry	Friday 10:00 AM	4123	Jason
16	7	Parnell	Friday 08:00 PM	2289	Tahlia

This table contains the following non-trivial functional dependencies:

{ID} → {BuildingNumber, BuildingName, DayTime, TutorNumber, TutorName}
{BuildingNumber} → {BuildingName}
{TutorNumber} → {TutorName}

Using the table above, you must provide a brief example and explanation of database operations which would cause an insertion, modification and deletion anomaly. Your explanation should be brief and not exceed 50 words. A dummy example of the format your answers should be written in can be seen on the next page.

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Example format:

Operation:
Insert <11, 7, 'Parnell', 'Monday 8:00 AM', 0007, 'Jack'> into Tutor

Explanation:
This operation would cause an insert anomaly to occur because …

Insertion Anomaly:

Operation:

Explanation:

Modification Anomaly:

Operation:

Explanation:

Deletion Anomaly:

Operation:

Explanation:

SEE NEXT PAGE FOR SECTION B

Section B – Keys

For each question in this section, you are required to list all possible candidate keys for the given schema based on the functional dependencies provided. You may wish to compute the closure of your key(s) to confirms they are valid.

Question 1

R [A, B, C, D, E, F, G, H, I, J]

{A} → {B}

{C} → {B}

{B} → {D, E, F, G, H}

{D, F} → {I, J, A}

Candidate Key(s):

Question 2

R [A, B, C, D, E, F, G, H]

{A, B} → {C, D, E}

{C} → {A, B, D, E}

{D} → {G}

{E} → {F}

Candidate Key(s):

SEE NEXT PAGE FOR SECTION C

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Section C – Highest Normal Form

For each question in this section, you are required state and justify the highest normal form of the relation given a schema and functional dependencies. Your explanations should be brief and concise. *Hint*: It may be useful to identify the candidate keys for each relation.

Question 1

R [A, B, C, D, E, F]

{B, C} → {D, E, F} 1NF

{A} → {D, E, F} 1NF

Highest Normal Form:

Explanation:

Question 2

R [A, B, C, D, E, F, G, H]

{A, B, C} → {D, E, F, G, H} BCNF

{G, H} → {A, B, C} 3NF

Highest Normal Form:

Explanation:

SEE NEXT PAGE FOR SECTION D

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Section D – BCNF Decomposition

For each question in this section, you are required to decompose the given relation into BCNF form and state any new relations created in the process with their functional dependencies and identify any functional dependencies which are lost during the decomposition. You must show your working using the tree method presented in tutorials. Consider the functional dependencies in the order presented in the question.

Question 1

R [A, B, C, D, E, F, G, H, I, J]

{A} → {B, C} BCNF

{B} → {D, E, F} 2NF

{C} → {G, H, I} 2NF

{H, I} → {F, J} 2NF

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Question 2

R [A, B, C, D, E, F, G, H]

{A, B, C} → {D, E, F, G} BCNF

{G, H} → {A, B, C} BCNF

{C} → {H} 3NF

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Section E – 3NF Decomposition

Question 1

Based on the following relational schema and functional dependencies, find minimal cover for relation R.

R [A, B, C, D, E, F, G, H, I, J, K, L, M, N]

{A} → {C, D, F, G}

{B} → {E}

{A, G} → {J, C}

{D, E, B} → {H, I, J}

{J} → {K, L, M}

{M} → {N}

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Question 2

The minimal cover has been provided below for a given relation with a set of functional dependencies. Using the minimal cover, normalise the relation to 3NF such that all functional dependencies are preserved.

R [A, B, C, D, E, F, G, H]

{A} → {D, F}

{B} → {G, E}

{F, G} → {H}

Minimal Cover: {

{A} → {D}, {B} → {G}, {B} → {E}, {A} → {F}, {F, G} → {H}

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